



**Universidad
Continental**

METODO DE COEFICIENTES ACI EN VIGAS Y LOSAS

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ANÁLISIS ESTRUCTURAL II

INGENIERIA CIVIL - FILIAL CUSCO
2022 -II



Restricción de análisis

El criterio que se debe tener para aplicar el método es que la luz mayor entre la luz menor sea igual a 1.2, es decir la luz mayor no exceda en 20% a la luz menor.

$$\frac{L_M}{L_m} = 1.2$$



Restricción de análisis

Vigas continuas, losas aligeradas o losas armadas en un solo sentido

Debe haber cargas **uniformemente distribuidas**; las cargas muertas y vivas uniformemente distribuidas en cada uno de los tramos **tienen la misma intensidad**

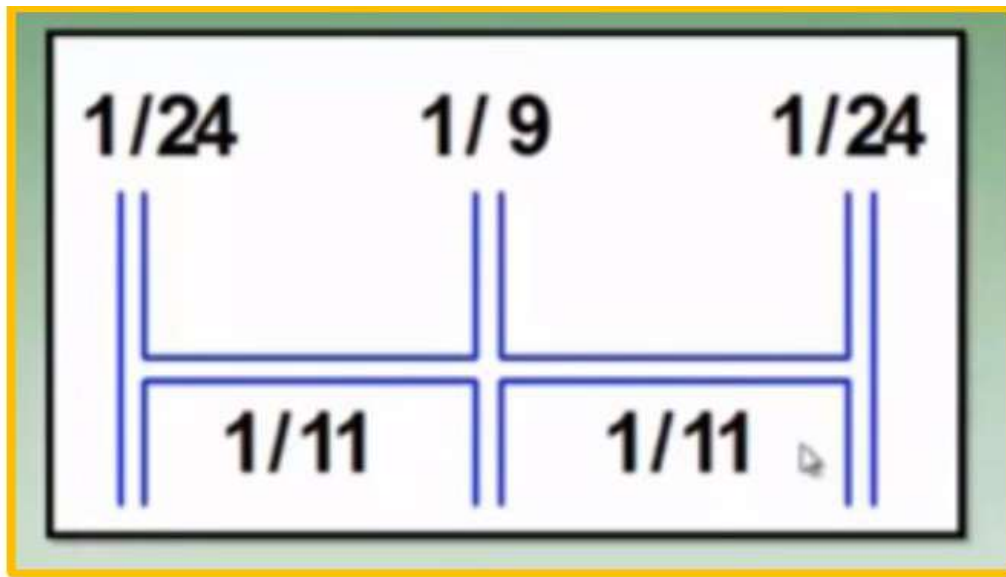


Restricción de análisis

La carga viva no debe exceder de **tres veces** la carga muerta

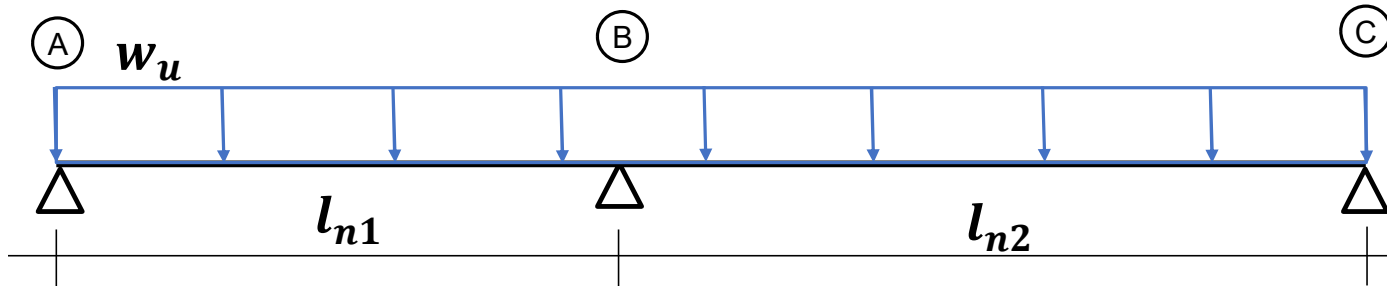


Caso 1: Apoyo Simple (Losa Aligerada)



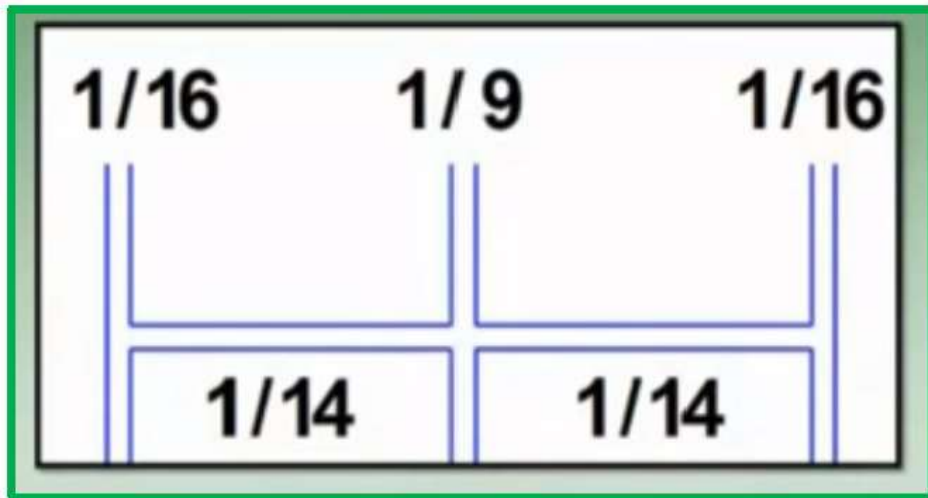
$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



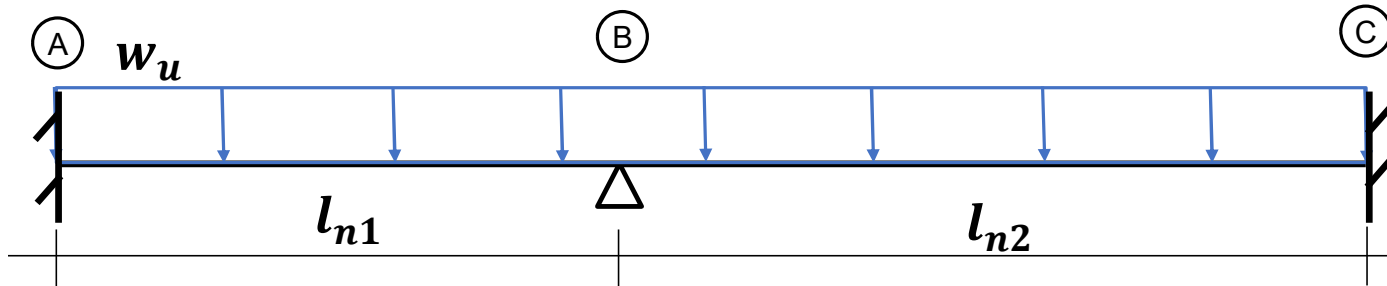


Caso 2 : Apoyo Empotrado (Vigas)



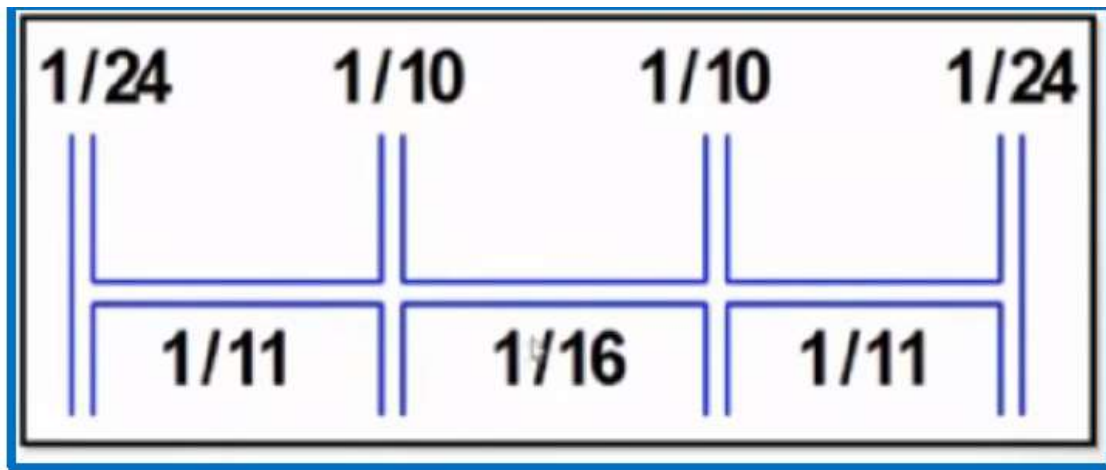
$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



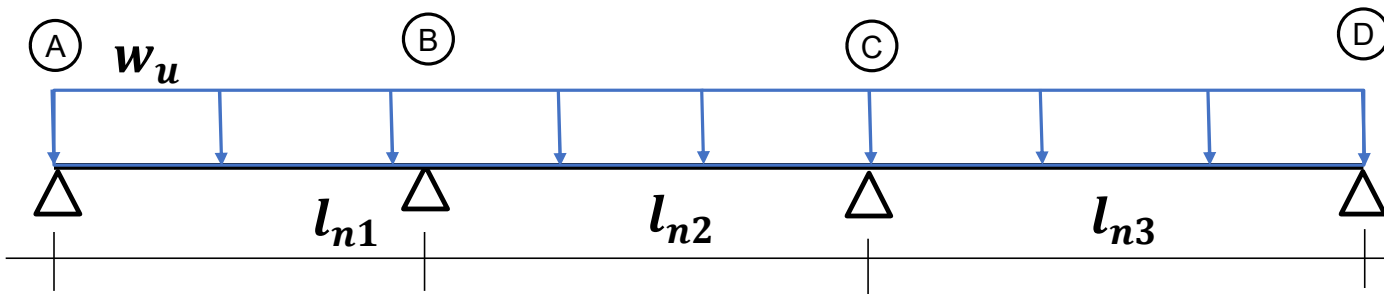


Caso 3 : Apoyo Simple (Losa Aligerada) Apoyo Empotrado



$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



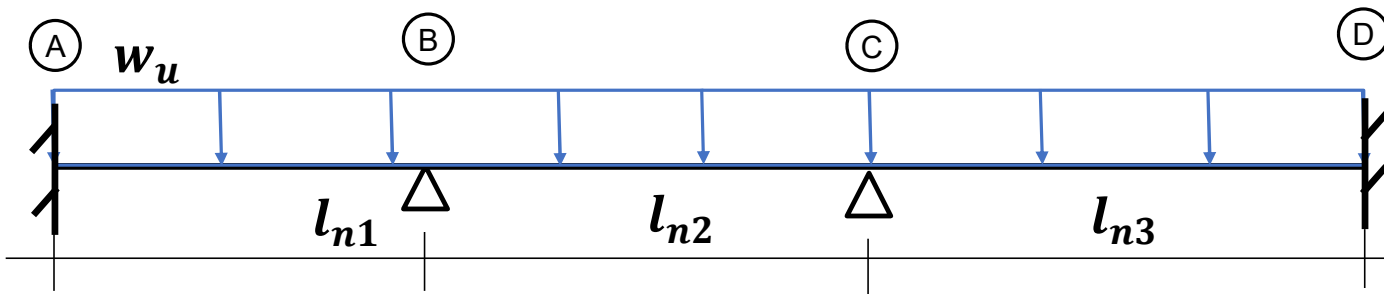


Caso 4 : Apoyo Empotrado (Vigas)



$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



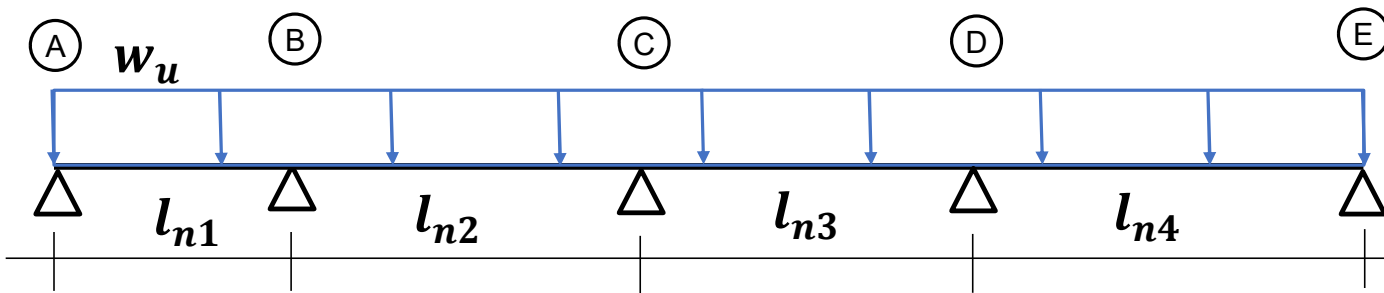


Caso 5 : Apoyo Simple (Losa Aligerada) Apoyo Empotrado

1/24	1/10	1/11	1/10	1/24
1/11	1/16	1/16	1/11	

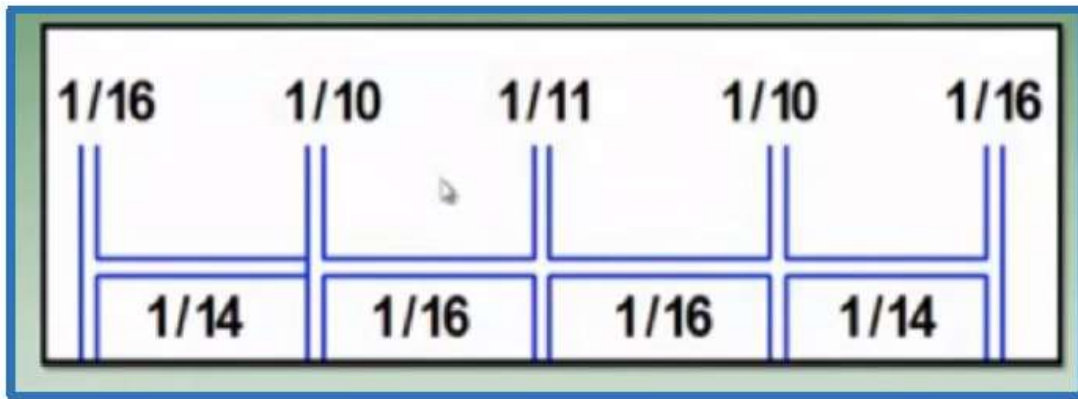
$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



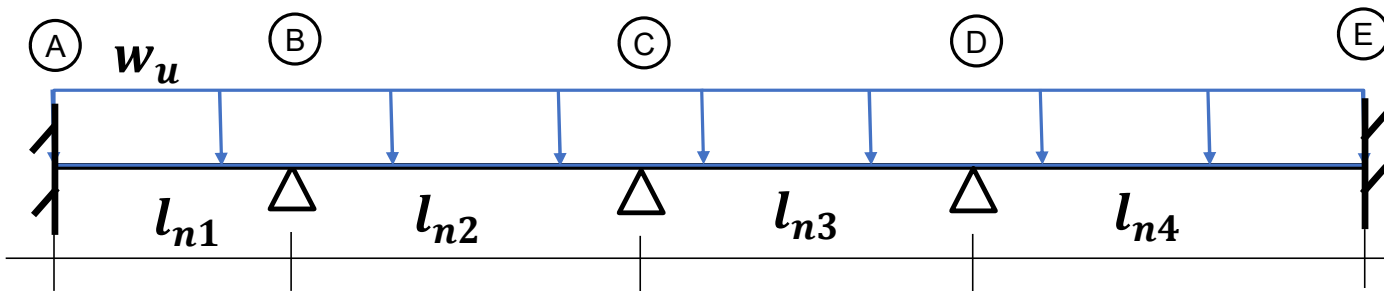


Caso 6 : Apoyo Empotrado (Vigas)



$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



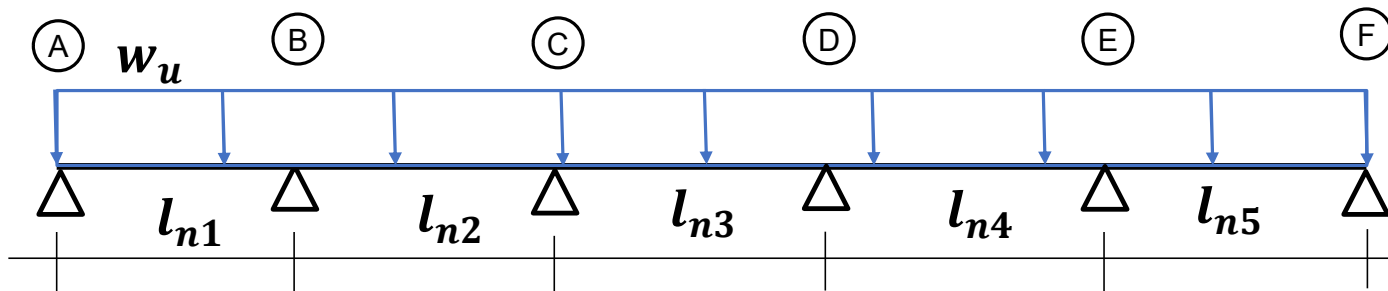


Caso 7 : Apoyo Simple (Losa Aligerada)

1/24	1/10	1/11	1/11	1/10	1/24
1/11	1/16	1/16	1/16	1/11	

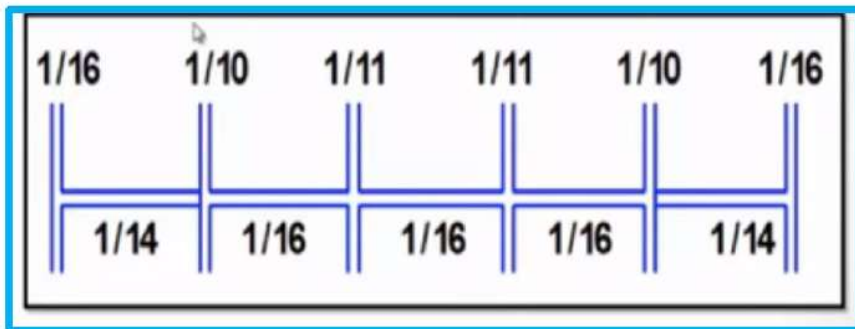
$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



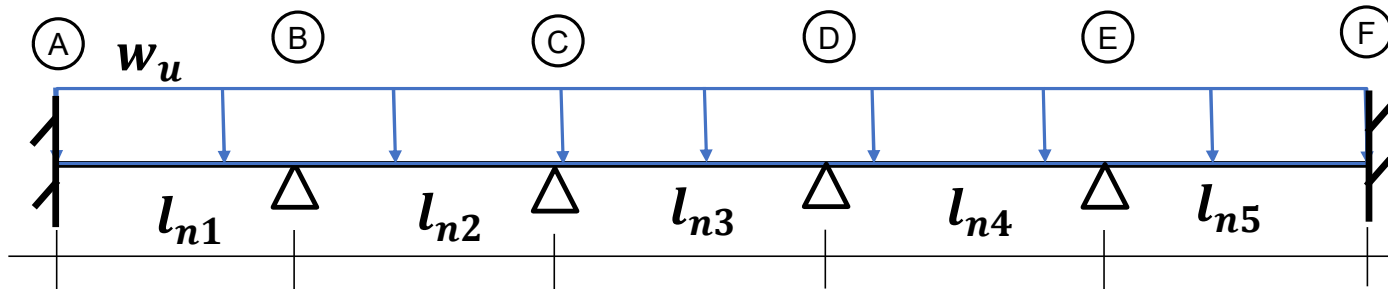


Caso 8 : Apoyo Empotrado (Vigas)



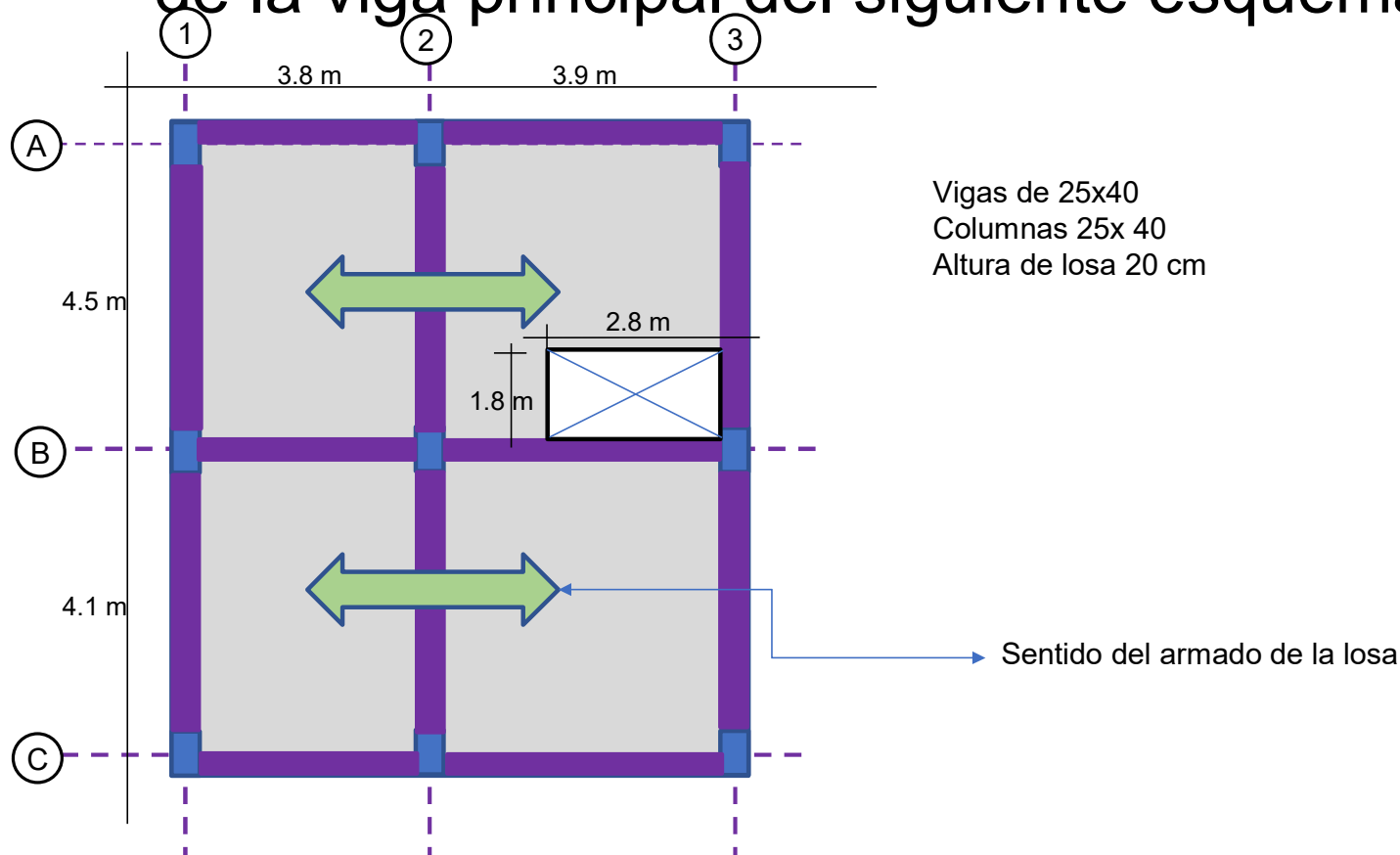
$$w_u l^2$$

$$w_u = 1.4w_m + 1.7w_v$$



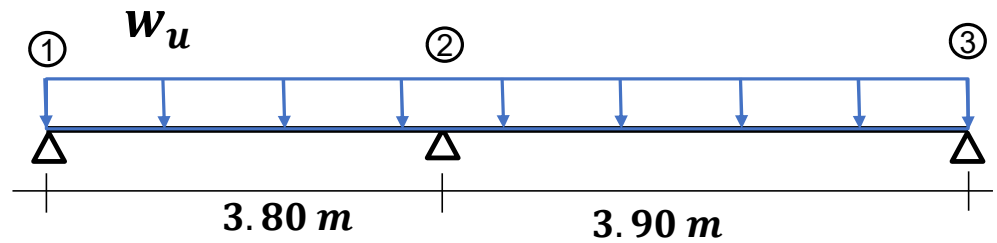
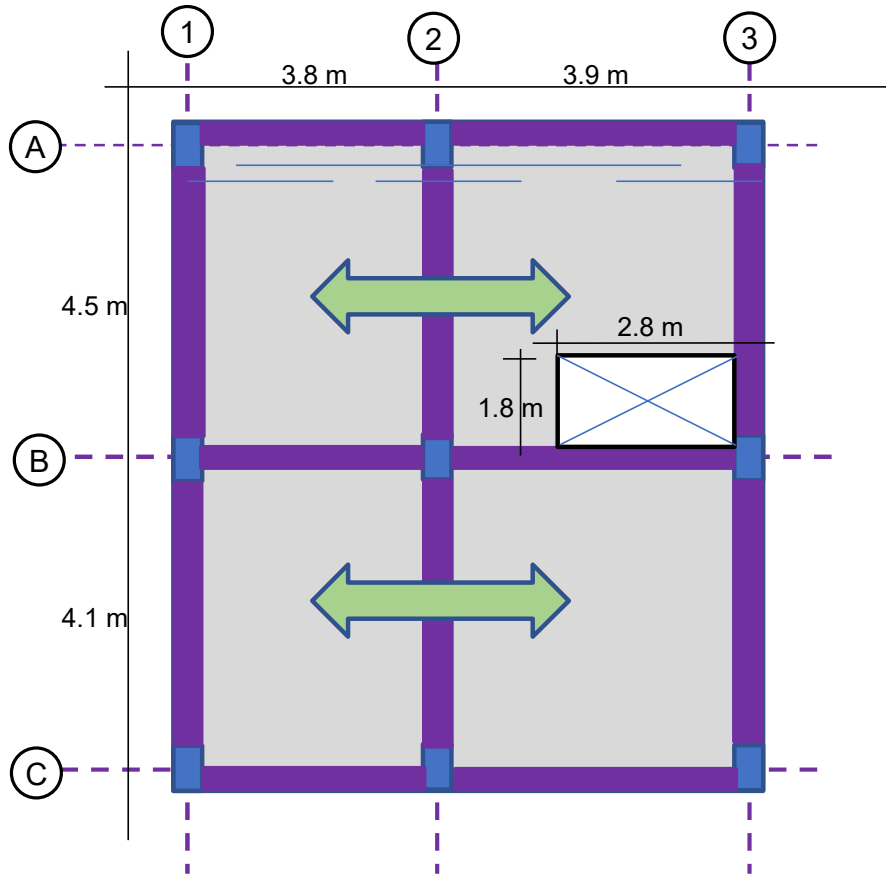


Ejemplo 1: Determinar los momentos de la losa y de la viga principal del siguiente esquema





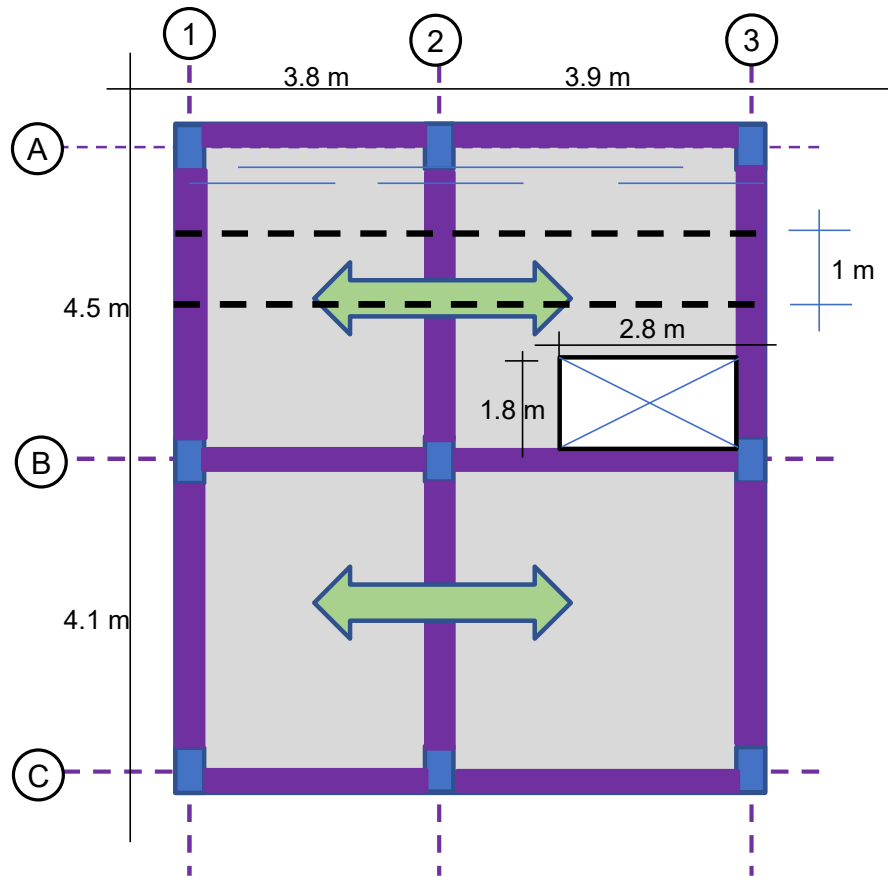
Análisis de losa : Verificación de restricción



$$\frac{L_M}{L_M} = \frac{3.90}{3.80} = 1.03 \leq 1.20$$



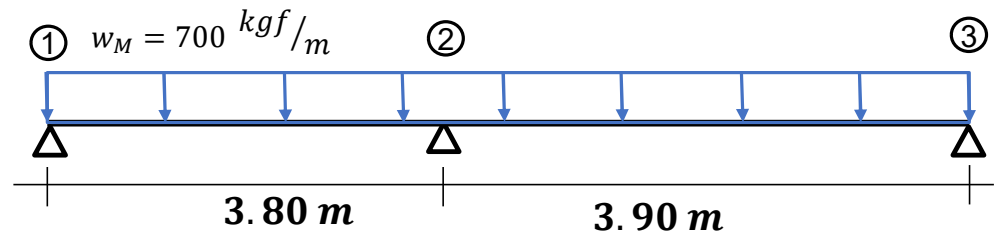
Análisis de losa : Metrado de cargas muertas



$$w_{PPL} = (1 \text{ m}) \left(300 \frac{\text{kgf}}{\text{m}^2} \right) = 300 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso propio de losa})$$

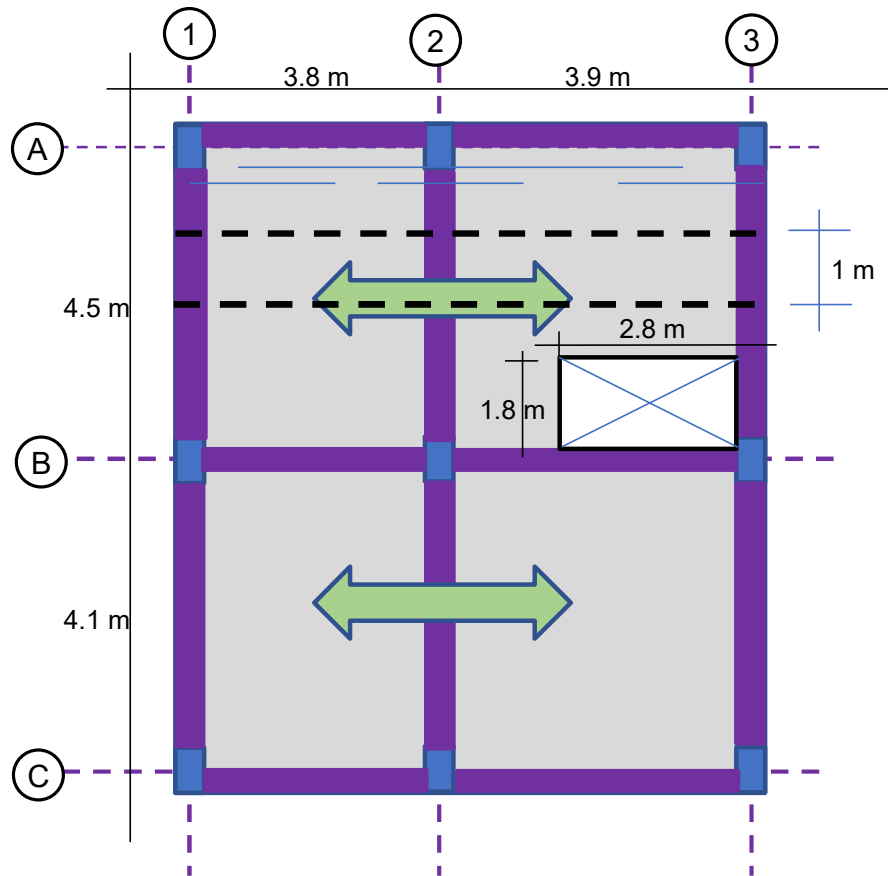
$$w_{PPT} = (1 \text{ m}) \left(400 \frac{\text{kgf}}{\text{m}^2} \right) = 400 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso propio de tabiquería})$$

$$w_M = w_{PPL} + w_{PPT} = 300 + 400 \frac{\text{kgf}}{\text{m}} = 700 \frac{\text{kgf}}{\text{m}}$$

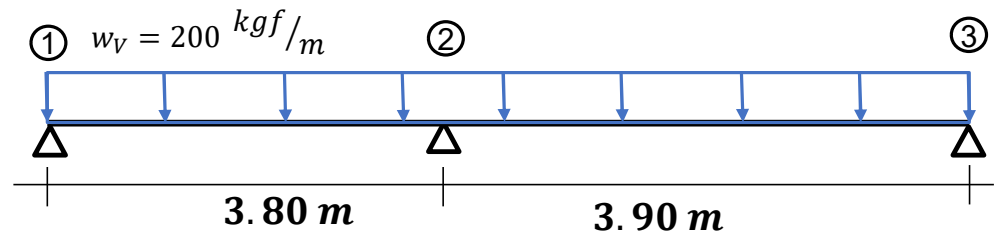




Análisis de losa : Metrado de cargas vivas

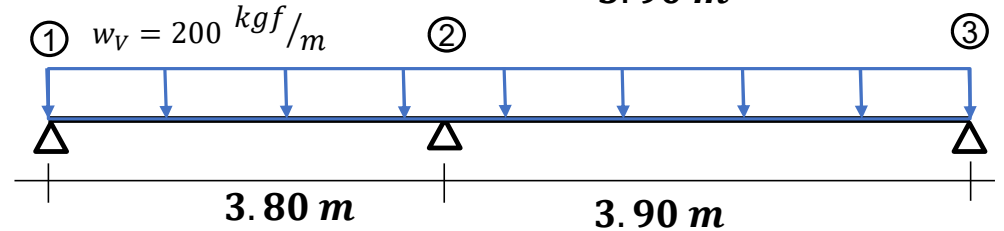
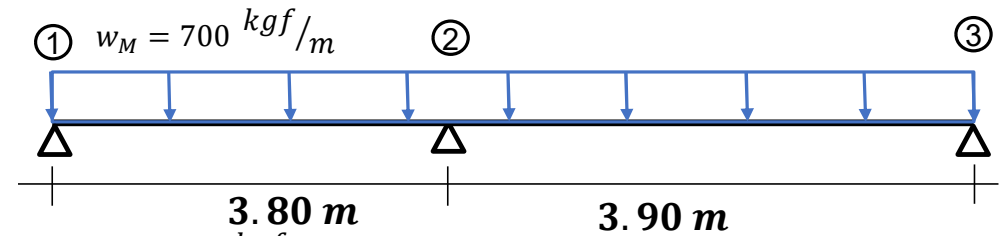
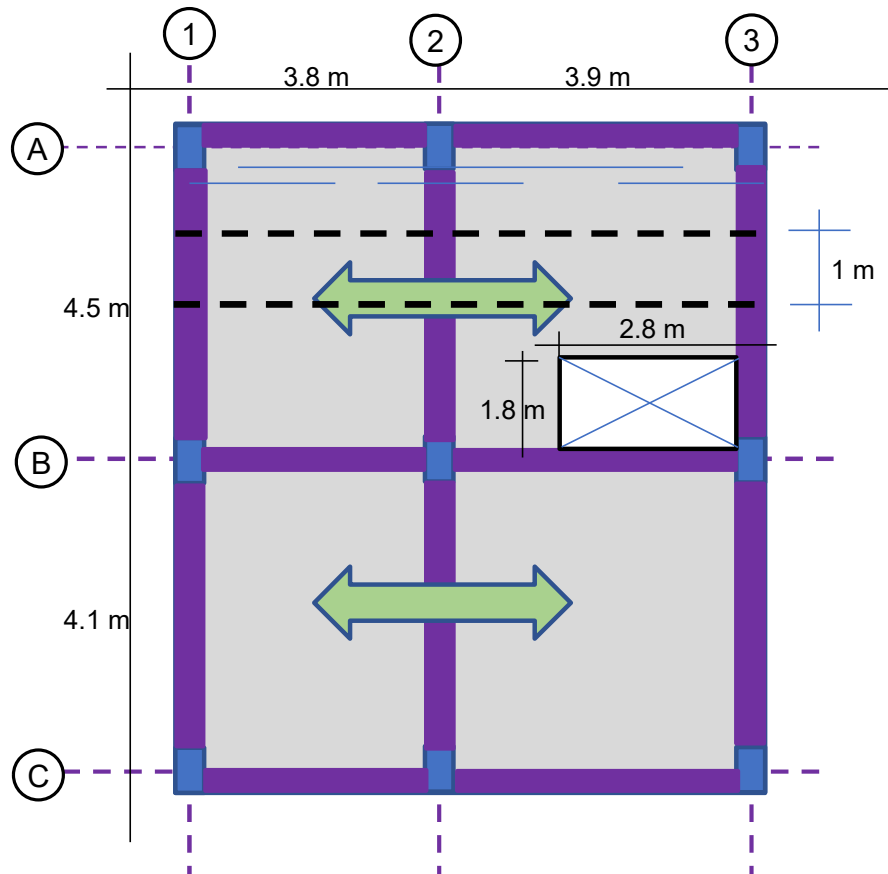


$$w_V = (1 \text{ m}) \left(200 \frac{\text{kgf}}{\text{m}^2} \right) = 200 \frac{\text{kgf}}{\text{m}} \quad (\text{Carga viva de vivienda})$$

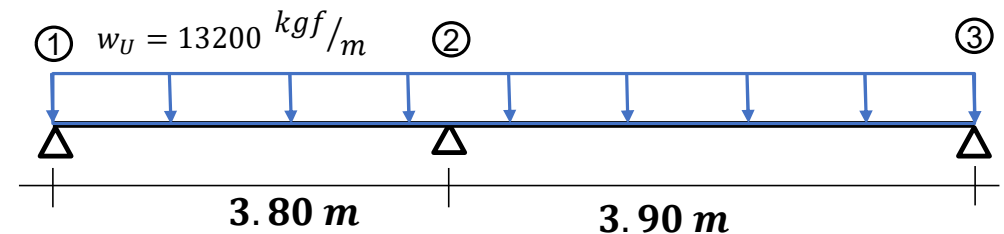




Análisis de losa : Determinación de la carga ultima

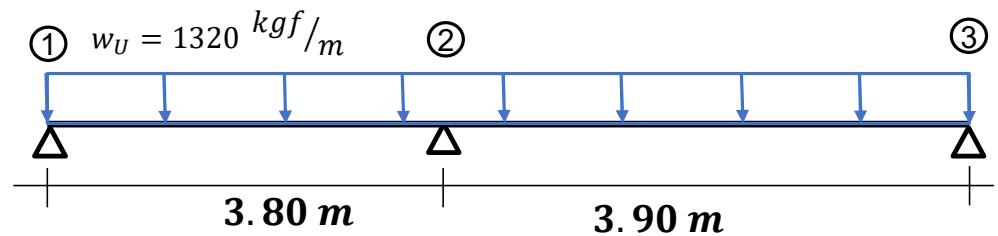
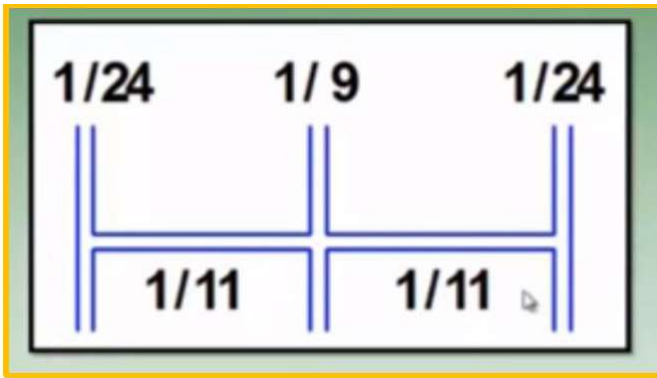


$$w_u = 1.4(700) + 1.7(200) = 1320 \text{ Kgf/m}$$





Análisis de losa : Determinación de los momentos utilizando el Método de Coeficientes del ACI



$$w_u l^2$$

$$M_1 = \frac{W_U l^2}{24} = \frac{(1320)(3.8)^2}{24} = 794.20 \text{ Kgf.m}$$

$$M_2 = \frac{W_U l^2}{11} = \frac{(1320)(3.8)^2}{11} = 1732.80 \text{ Kgf.m}$$

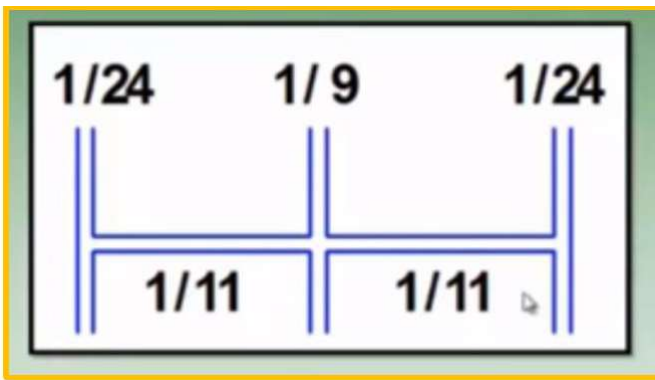
$$M_5 = \frac{W_U l^2}{24} = \frac{(1320)(3.9)^2}{24} = 836.55 \text{ Kgf.m}$$

$$M_3 = \frac{W_U l^2}{9} = \frac{(1320) \left(\frac{3.8 + 3.9}{2} \right)^2}{9} = 2173.96 \text{ Kgf.m}$$

$$M_4 = \frac{W_U l^2}{11} = \frac{(1320)(3.9)^2}{11} = 1825.20 \text{ Kgf.m}$$



Análisis de losa : Determinación de los momentos utilizando el Método de Coeficientes del ACI



$$w_u l^2$$

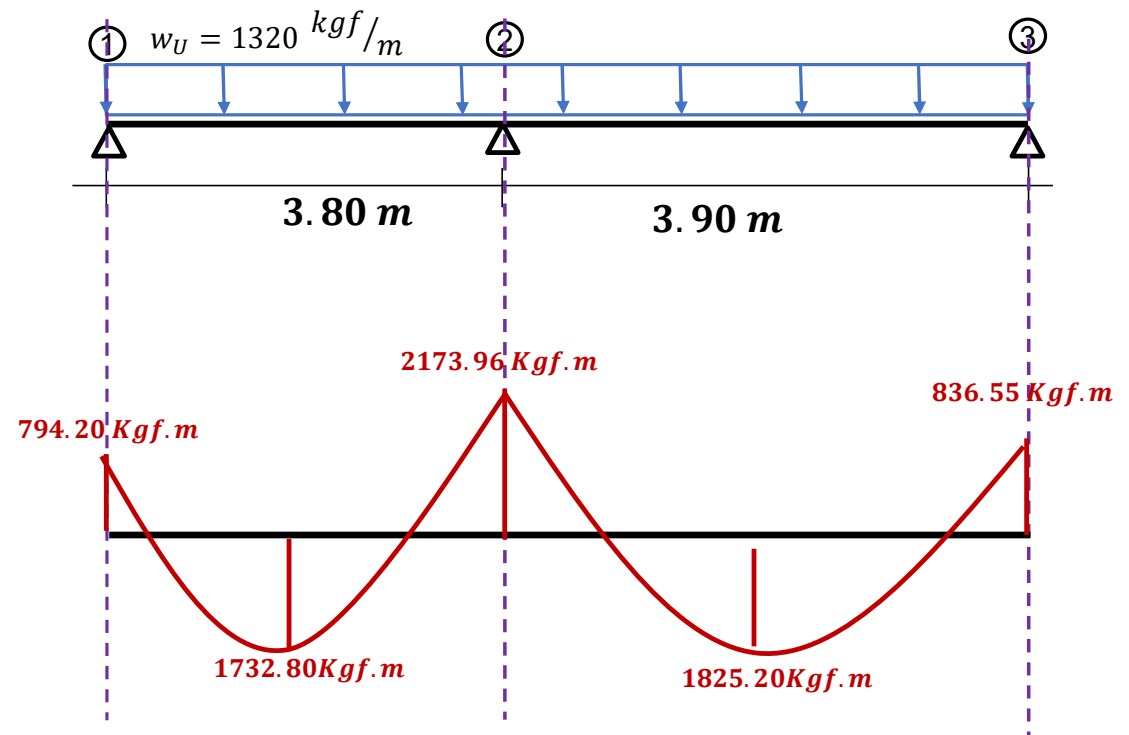
$$M_1 = 794.20 \text{ Kgf.m}$$

$$M_2 = 1732.80 \text{ Kgf.m}$$

$$M_3 = 2173.96 \text{ Kgf.m}$$

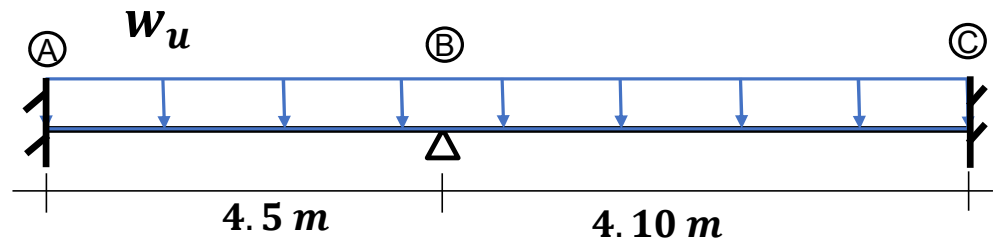
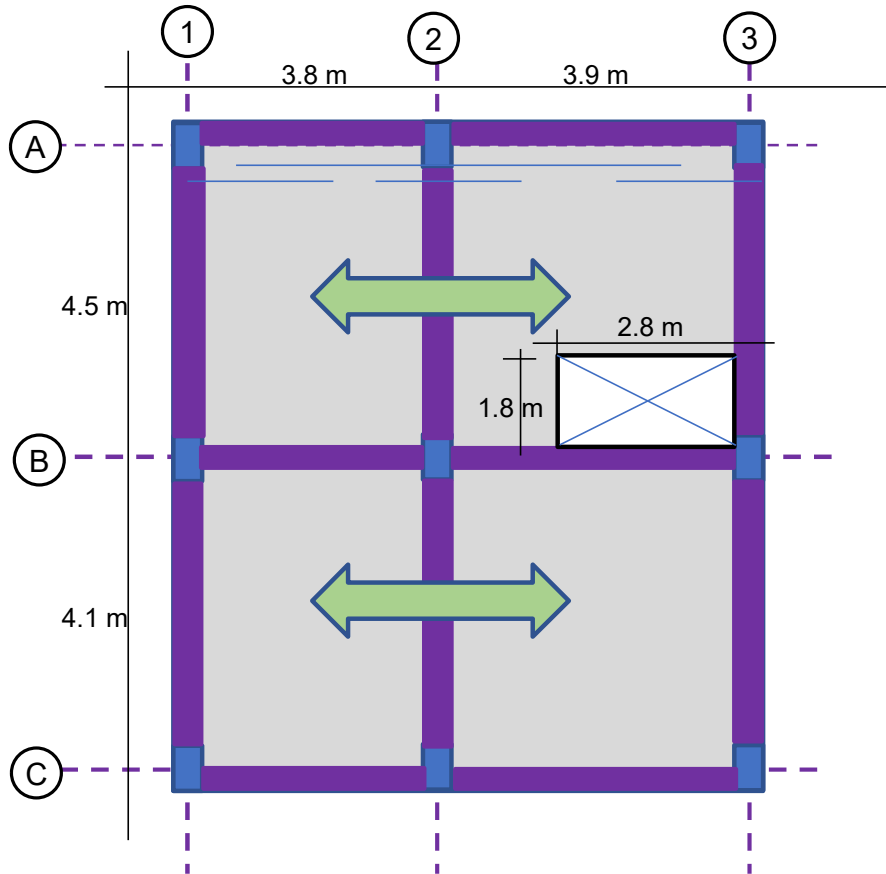
$$M_4 = 1825.20 \text{ Kgf.m}$$

$$M_5 = 836.55 \text{ Kgf.m}$$





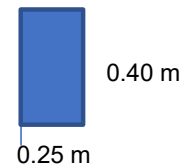
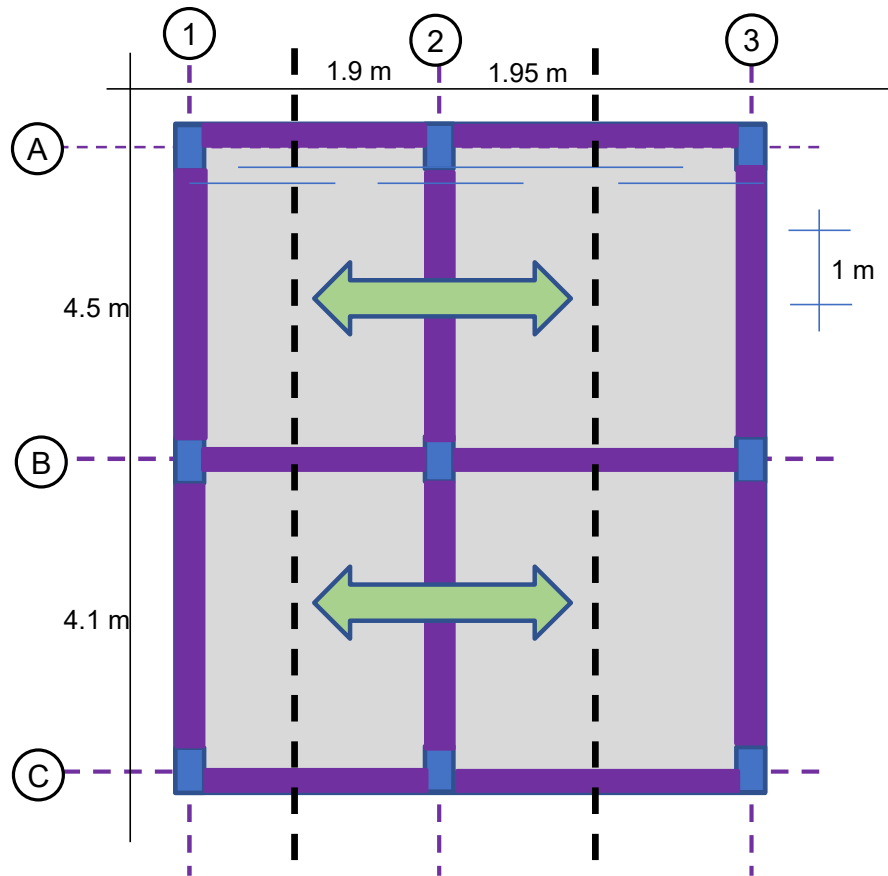
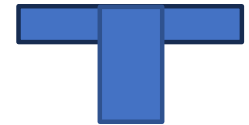
Análisis de viga eje 2 : Verificación de restricción



$$\frac{L_M}{L_m} = \frac{4.50}{4.10} = 1.10 \leq 1.20$$



Análisis de vigas : Metrado de cargas muertas



$$w_{PPV} = (0.25 \text{ m})(0.40 \text{ m}) \left(2400 \frac{\text{kgf}}{\text{m}^3} \right) = 240 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso propio de viga})$$

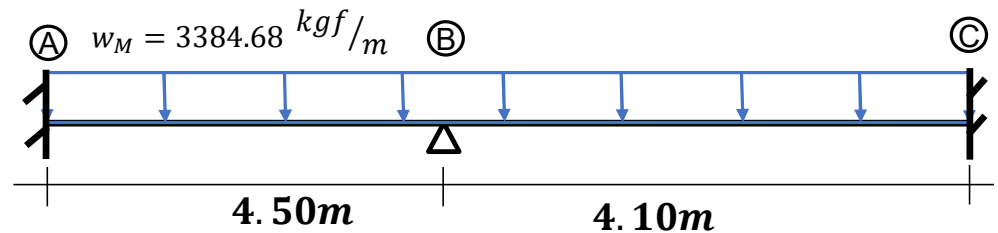
$$w_{PPL} = (1.90 + 1.95 - 0.25) \left(300 \frac{\text{kgf}}{\text{m}^2} \right) = 1080 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso propio de losa})$$

$$w_{PPT} = (1.90 + 1.95 \text{ m}) \left(400 \frac{\text{kgf}}{\text{m}^2} \right) = 1540 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso propio de tabiquería})$$

$$w_{PPEt} = (1.9 + 1.95 \text{ m}) (0.0254 \text{ m}) \left(1000 \frac{\text{kgf}}{\text{m}^3} \right) = 97.79 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso enlucado})$$

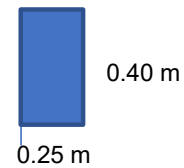
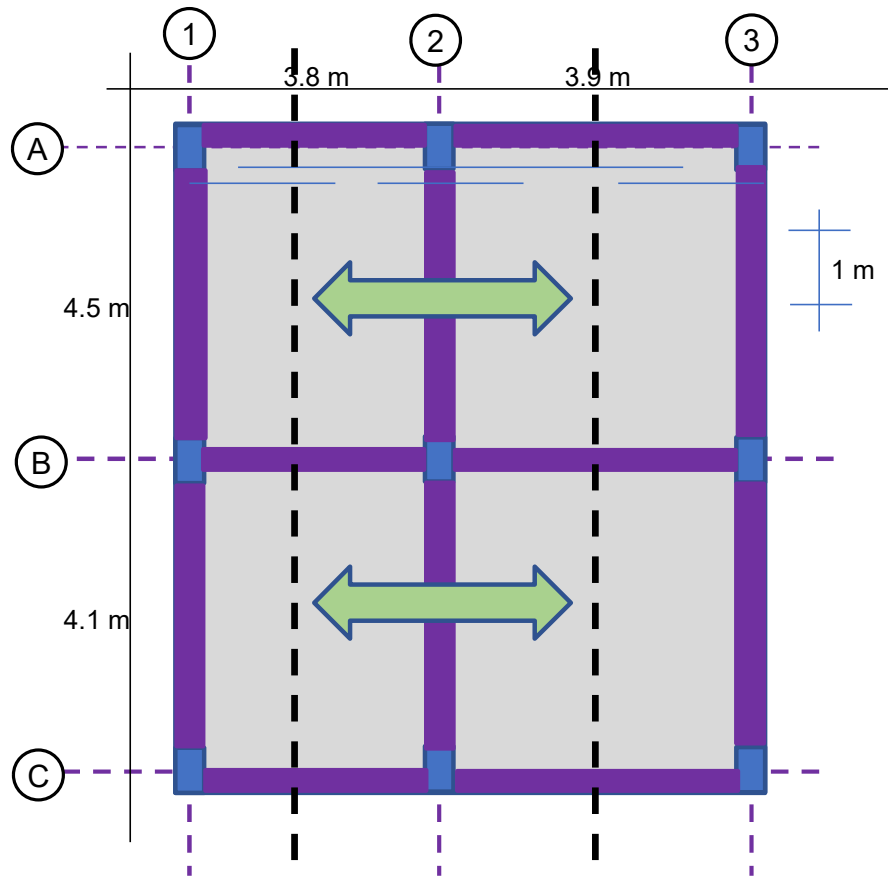
$$w_{PPCP} = (1.9 + 1.95) (0.0504 \text{ m}) \left(2200 \frac{\text{kgf}}{\text{m}^3} \right) = 426.89 \frac{\text{kgf}}{\text{m}} \quad (\text{Peso contrafrío})$$

$$w_M = 240 + 1080 + 1540 + 97.79 + 426.89 = 3384.68 \frac{\text{kgf}}{\text{m}}$$

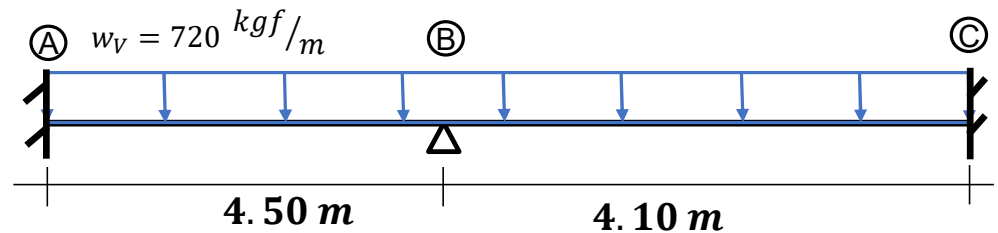




Análisis de vigas : Metrado de cargas viva

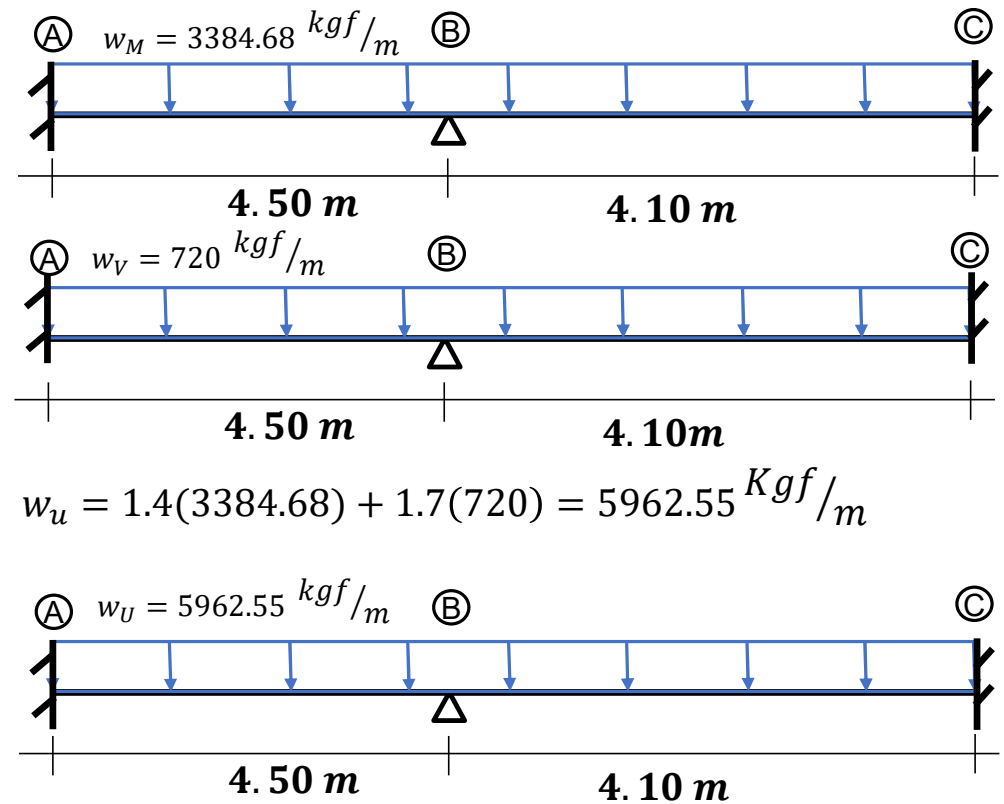
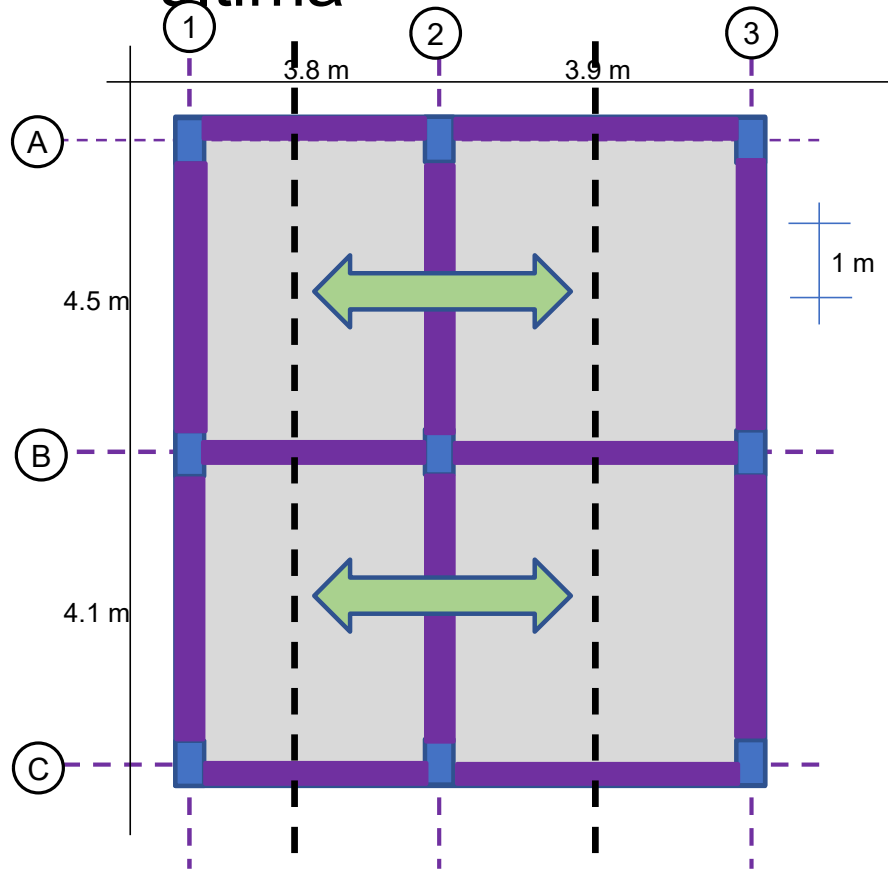


$$w_V = (1.9 + 1.85 - 0.25) (200 \text{ kgf/m}^2) = 720 \text{ kgf/m} \text{ (Carga viva)}$$





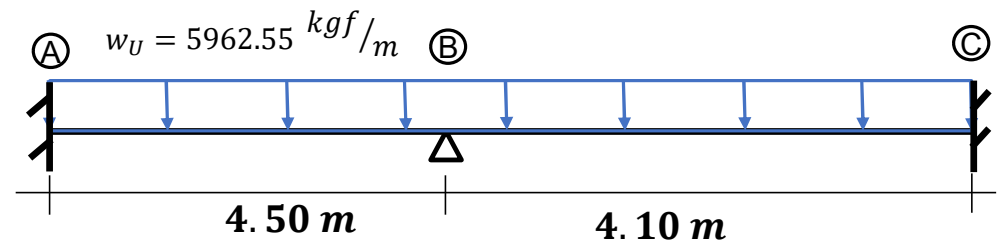
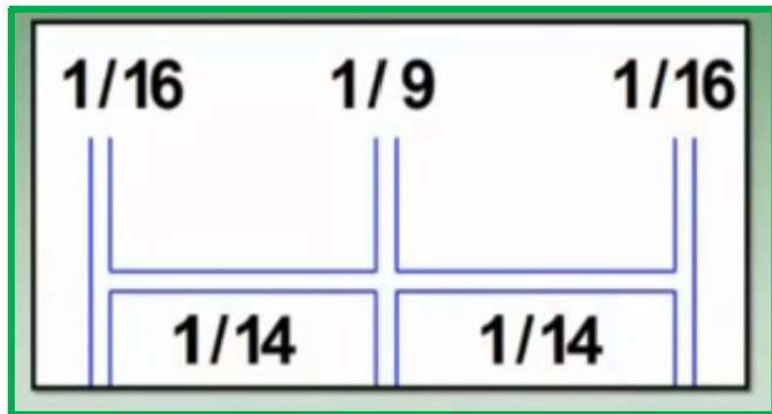
Análisis de viga 2 : Determinación de la carga ultima



$$w_u = 1.4(3384.68) + 1.7(720) = 5962.55 \text{ kgf/m}$$



Análisis de losa : Determinación de los momentos utilizando el Método de Coeficientes del ACI



$$M_1 = \frac{W_U l^2}{16} = \frac{(5962.55)(4.5)^2}{16} = 7546.35 \text{ Kgf.m}$$

$$M_2 = \frac{W_U l^2}{14} = \frac{(5962.55)(4.5)^2}{14} = 8624.40 \text{ Kgf.m}$$

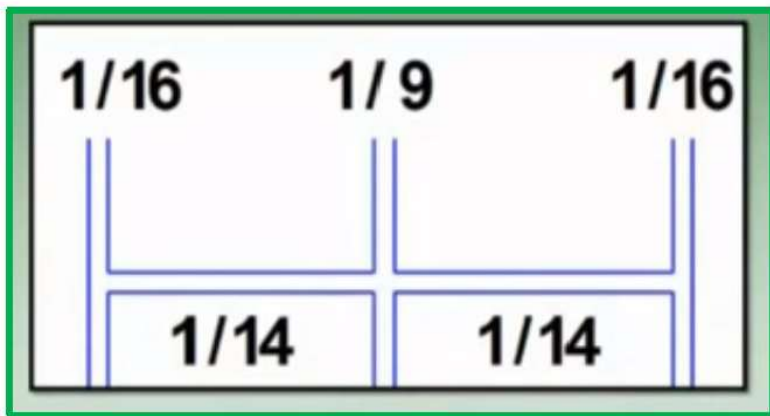
$$M_5 = \frac{W_U l^2}{16} = \frac{(5962.55)(4.1)^2}{16} = 6264.40 \text{ Kgf.m}$$

$$M_3 = \frac{W_U l^2}{9} = \frac{(5962.55) \left(\frac{4.50 + 4.10}{2} \right)^2}{9} = 12249.73 \text{ Kgf.m}$$

$$M_4 = \frac{W_U l^2}{14} = \frac{(5962.55)(4.1)^2}{14} = 7159.32 \text{ Kgf.m}$$



Análisis de viga : Determinación de los momentos utilizando el Método de Coeficientes del ACI



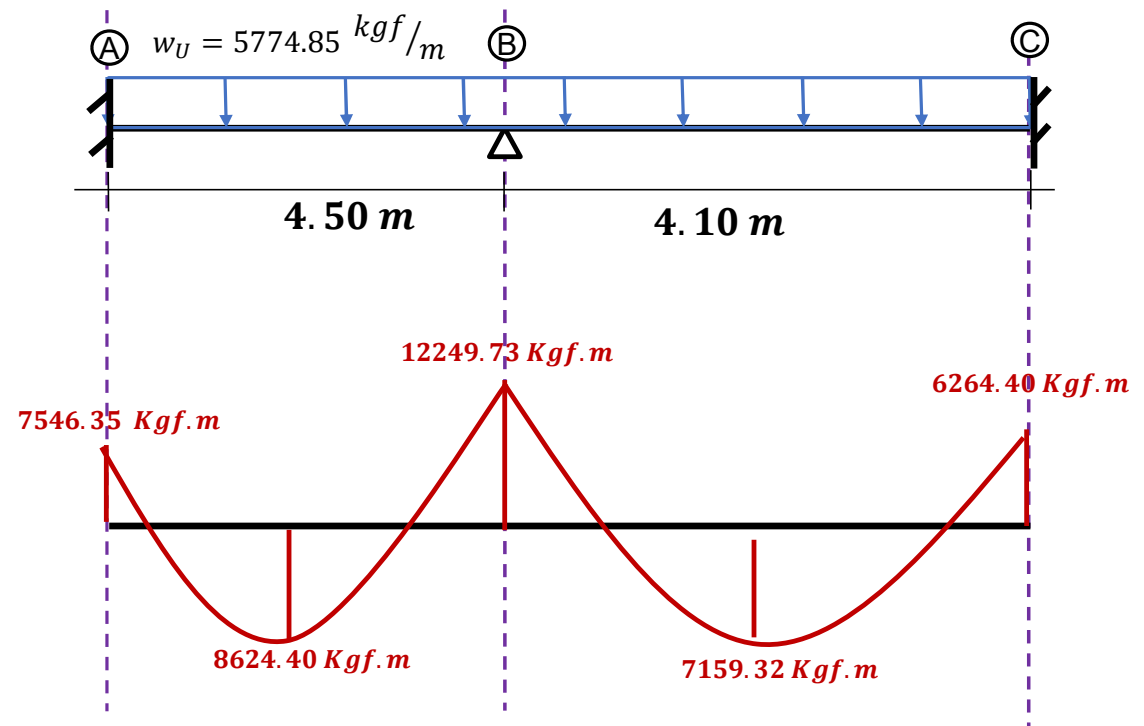
$$M_1 = 7546.35 \text{ Kgf.m}$$

$$M_2 = 8624.40 \text{ Kgf.m}$$

$$M_3 = 12249.73 \text{ Kgf.m}$$

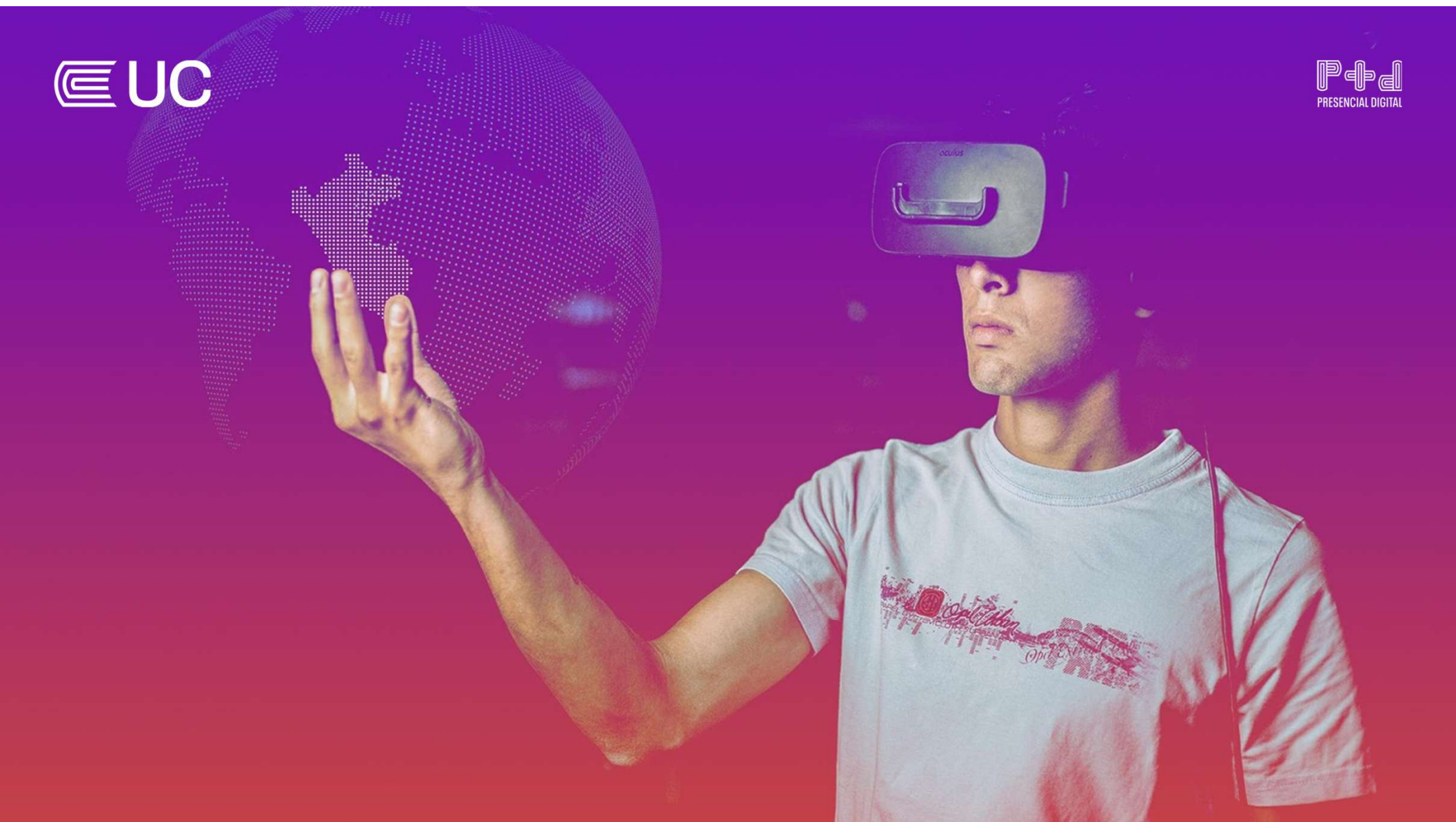
$$M_4 = 7159.32 \text{ Kgf.m}$$

$$M_5 = 6264.40 \text{ Kgf.m}$$



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